ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **How do Organisms Reproduce?** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** produce new individuals like themselves	Basic definition.
2	**(c)** pollen grains	Anther is the pollen-making part.
3	**(c)** sperm	Sperm cells are male gametes.
4	**(c)** binary fission	Unicellular organisms divide.
5	**(c)** stigma	It receives pollen.
6	**(b)** ovum	Egg cell.
7	**(c)** budding	Small bulges grow and detach.
8	**(a)** fruit	After fertilisation.
9	**(b)** testes	The male hormone.
10	**(b)** sporangia	Sporangia are spore-bearing structures.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Production of new individuals of the same kind (2).	2
12	Tiny reproductive units (1); Covered by thick protective walls (1).	1+1
13	Fusion of male gamete and female gamete (1); Forms the zygote (1).	1+1
14	Amoeba (1); Paramecium or Leishmania (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Binary fission: Amoeba (1); Budding: Hydra (1); Spore formation: Rhizopus (1).	1+1+1
16	Ovary (1); Oviduct (1); Uterus (1).	1+1+1
17	Plants bear flowers and fruits earlier (1); Seedless plants can be grown (1); Plants are genetically identical to the parent (1).	1+1+1
	OR: Sexual: two parents; asexual: one parent (1); Sexual: gametes involved; asexual: no gametes (1); Sexual: more variation; asexual: little variation (1).	1+1+1
18	Produce sperms (1); Secrete testosterone (1); Located in the scrotum for a lower temperature (1).	1+1+1
	OR: Only one organism is involved (1); Pieces grow into new individuals (1); No gametes are needed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Amoeba is a single cell with a nucleus (1); The nucleus divides into two (1); Cytoplasm constricts in the middle (1); Two daughter cells separate (1); Each grows into a new Amoeba (1).	1+1+1+1+1
	OR: Pollen lands on the stigma (1); It grows a pollen tube through the style (1); The tube reaches the ovule in the ovary (1); Male gamete fuses with the female gamete (1); The zygote is formed (1).	1+1+1+1+1
20	A small bud forms due to repeated cell division (1); The bud grows in size (1); It develops mouth and tentacles (1); It separates from the parent (1); It lives independently (1).	1+1+1+1+1
	OR: Testes produce sperms (1); Scrotum holds the testes (1); Vas deferens carries sperms (1); Prostate and seminal vesicle add fluid (1); Urethra through the penis carries semen out (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that Plasmodium reproduces by binary fission; it undergoes **multiple** fission, whereas Amoeba shows binary fission.
- Using 'fertilisation' for pollination; pollination is transfer of pollen, fertilisation is fusion of male and female gametes.
- Saying that fertilisation takes place in the uterus; it occurs in the oviduct and the zygote is implanted in the uterus.
- Confusing regeneration with reproduction by fragmentation; regeneration uses specialised cells to rebuild a missing part, e.g. Planaria, Hydra.
- Stating that asexual reproduction produces absolutely identical offspring; tiny variations are produced by errors in DNA copying.

- Believing that the testes lie inside the abdomen; they are in the scrotum because sperm formation requires a lower temperature.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **How do Organisms Reproduce?** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** seed	Fertilised ovule is a seed.
2	**(b)** uterus	Womb.
3	**(d)** regeneration	It regrows lost parts.
4	**(c)** brightly coloured	They attract pollinators.
5	**(a)** oviduct	Fallopian tube.
6	**(a)** one parent	Only one parent is needed.
7	**(b)** bisexual	Hibiscus is an example.
8	**(d)** placenta	Mother's blood supplies nutrients.
9	**(d)** Fertilisation	Fertilisation needs gametes.
10	**(b)** wind, insects and water	Pollinating agents.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Anther (1); Filament (1).	1+1
12	Male: testes (1); Female: ovaries (1).	1+1
13	Budding (2).	2
14	Transfer of pollen grains from anther to stigma (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sepal (1); Petal and stamen (1); Pistil (1).	1+1+1
16	Boys: beard and moustache (1); Boys: voice becomes deeper (1); Girls: breast growth and menstruation (1).	1+1+1
17	A reproductive cell that fuses during fertilisation (1); Sperm (1); Ovum (egg) (1).	1+1+1
	OR: Attracts pollinators with colour and smell (1); Produces pollen and ovules (1); Forms fruit and seeds after fertilisation (1).	1+1+1
18	Self-pollination: pollen reaches the stigma of the same plant (1); Cross-pollination: pollen goes to another plant (1); Both need a pollinating agent (1).	1+1+1
	OR: It divides into many daughter cells at once (1); The nucleus divides repeatedly first (1); Many cells form inside a cyst (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hyphae grow on bread (1); Upright hyphae bear sporangia (1); Sporangia hold spores (1); Spores are released and carried by air (1); They germinate when they land on a moist surface (1).	1+1+1+1+1
	OR: Ovaries produce eggs (1); Oviduct carries the egg (1); Uterus holds the developing baby (1); Cervix connects uterus to vagina (1); Vagina is the birth canal (1).	1+1+1+1+1
20	Regrowth of missing body parts (1); Planaria cut in pieces (1); Each piece grows into a new Planaria (1); Specialised cells produce the new parts (1); Not all organisms can regenerate (1).	1+1+1+1+1
	OR: Zygote forms in the oviduct (1); Divides repeatedly into an embryo (1); Embryo is implanted in the uterus (1); Placenta provides nutrition (1); Foetus grows for about nine months (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that Plasmodium reproduces by binary fission; it undergoes **multiple** fission, whereas Amoeba shows binary fission.
- Using 'fertilisation' for pollination; pollination is transfer of pollen, fertilisation is fusion of male and female gametes.
- Saying that fertilisation takes place in the uterus; it occurs in the oviduct and the zygote is implanted in the uterus.
- Confusing regeneration with reproduction by fragmentation; regeneration uses specialised cells to rebuild a missing part, e.g. Planaria, Hydra.

- Stating that asexual reproduction produces absolutely identical offspring; tiny variations are produced by errors in DNA copying.
- Believing that the testes lie inside the abdomen; they are in the scrotum because sperm formation requires a lower temperature.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **How do Organisms Reproduce?** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** sexual maturity begins	Reproductive organs mature.
2	**(c)** Bryophyllum	Plantlets arise on leaf margins.
3	**(a)** seed coat	Outer covering.
4	**(c)** epididymis and vas deferens	Male storage and transport.
5	**(c)** grow into a new plant	Stem cuttings root.
6	**(b)** zygote	Zygote is the first cell of the new organism.
7	**(a)** one egg	Usually one egg.
8	**(a)** DNA copying	DNA carries the instructions.
9	**(c)** the micropyle	A tiny opening in the ovule.
10	**(b)** vagina	Birth canal.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Monthly bleeding from the uterus (1); Occurs if the egg is not fertilised (1).	1+1
12	An outgrowth of the body (1); It grows into a new Hydra (1).	1+1
13	Wind (1); Insects (or water) (1).	1+1
14	Gives nutrition and oxygen to the embryo and removes waste (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A flower with only stamens or only pistil (1); Papaya (1); Watermelon (1).	1+1+1
16	Sprouting of a seed into a seedling (1); Water (1); Air and warmth (1).	1+1+1
17	They wither (1); They fall off (1); Only the ovary continues to grow (1).	1+1+1
	OR: Testis (1); Vas deferens (1); Urethra (or prostate gland) (1).	1+1+1
18	The embryo cannot make food (1); Placenta transfers nutrients and oxygen (1); It removes waste products (1).	1+1+1
	OR: Embryo is implanted in its wall (1); It holds and nourishes the growing baby (1); Its muscles help in childbirth (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Stigma receives pollen (1); Style is the stalk joining stigma to the ovary (1); Pollen tube grows through the style (1); Ovary holds ovules (1); Ovule contains the egg cell (1).	1+1+1+1+1
	OR: Egg is released monthly (1); Uterine lining thickens (1); If egg is not fertilised the lining breaks down (1); Blood and tissues leave the body (1); The cycle repeats every month (1).	1+1+1+1+1
20	Transfer of pollen from anther to stigma (1); Self-pollination: same flower or same plant (1); Cross-pollination: different plants (1); Agents: wind, insects, water, animals (1); Cross-pollination gives variation (1).	1+1+1+1+1
	OR: Puberty begins around 10 to 14 years (1); Boys: facial hair, deeper voice, sperm production (1); Girls: breast development, menstruation (1); Hair appears in the underarm and genital area (1); Reproductive organs become functional (1).	1+1+1+1+1

Mistakes to avoid:

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- Using 'fertilisation' for pollination; pollination is transfer of pollen, fertilisation is fusion of male and female gametes.
- Saying that fertilisation takes place in the uterus; it occurs in the oviduct and the zygote is implanted in the uterus.
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